# Assignment-3

## MCA | Machine Learning

---

**Question 1 (5 marks)**

**Define Supervised Learning. Explain the key difference between supervised learning and unsupervised learning with one example of each.**

### 1. Definition

Supervised learning is a machine learning method where the model is trained on **labeled data**, i.e., every input (X) comes with the correct output (Y). The model learns the mapping **Y = f(X)** and then predicts Y for new, unseen inputs.

### 2. How It Works

Give data with answers → model predicts → compare with the correct label → reduce the error → repeat. It is like a student practising with an **answer key**.

### 3. Unsupervised Learning

The data has **no labels** (only X). The model finds hidden patterns, groups or structure on its own. There is no answer key.

### 4. Key Difference

| Basis | Supervised | Unsupervised |
|---|---|---|
| **Data** | Labeled (X and Y) | Unlabeled (only X) |
| **Goal** | Predict an output | Discover hidden structure |
| **Feedback** | Yes, errors are measured | No feedback |
| **Tasks** | Classification, Regression | Clustering, Association |
| **Algorithms** | Linear Regression, SVM | K-Means, PCA |

### 5. Example – Supervised

**Spam detection:** emails are labeled "Spam" or "Not Spam". The model learns from them and classifies new emails.

### 6. Example – Unsupervised

**Customer segmentation:** shopping data has no labels. K-Means groups customers into clusters such as budget, regular and premium buyers.

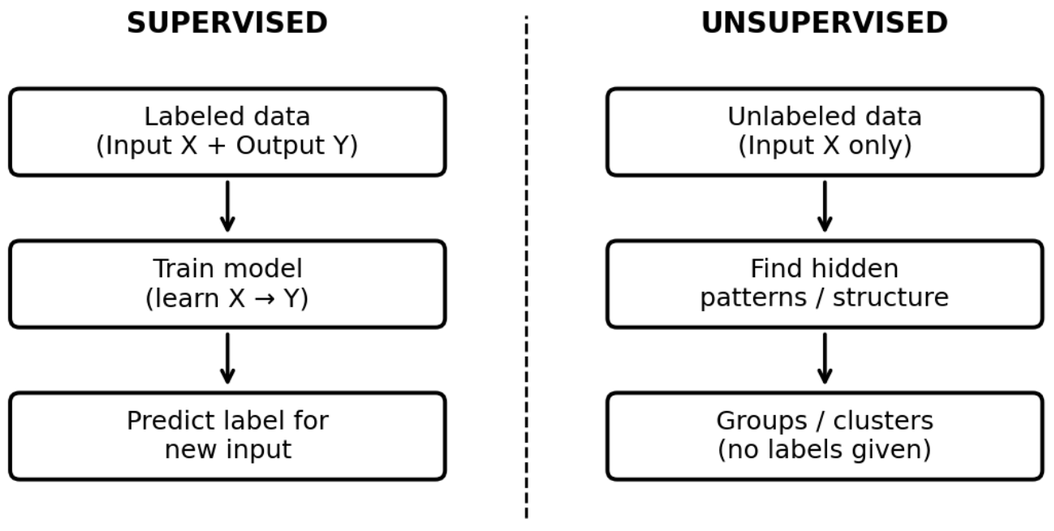



*Fig. 1: Supervised vs Unsupervised learning (flowchart)*

## Question 2 (5 marks)

**What are the two main types of supervised learning problems? Clearly distinguish between them and give one real-world example for each type.**

### 1. The Two Types

Supervised problems are divided by the **type of output** to predict: **Classification** and **Regression.**

### 2. Classification

Predicts a **discrete category/class label** from a fixed set (Yes/No, Spam/Not Spam, Cat/Dog/Bird). It can be binary or multi-class. Algorithms: Logistic Regression, SVM, Decision Tree.

### 3. Regression

Predicts a **continuous numeric value** that can take any value in a range (price, temperature, marks). Algorithms: Linear Regression, Random Forest Regressor.

### 4. Key Differences

| Basis | Classification | Regression |
|---|---|---|
| Output | Category (discrete) | Number (continuous) |
| Sample output | "Spam", "Malignant" | ₹45.6 lakh, 32.5 °C |
| Graph | A boundary separates classes | A best-fit line/curve through data |
| Metrics | Accuracy, Precision, Recall, F1 | MSE, RMSE, MAE, $R^2$ |

### 5. Real-World Examples

- **Classification:** tumor detection. Scan features → Benign or Malignant.
- **Regression:** house price prediction. Area, bedrooms, location → price in ₹.

### 6. Memory Trick

Ask: "Is the answer a **category** or a **number**?" Category → Classification. Number → Regression.

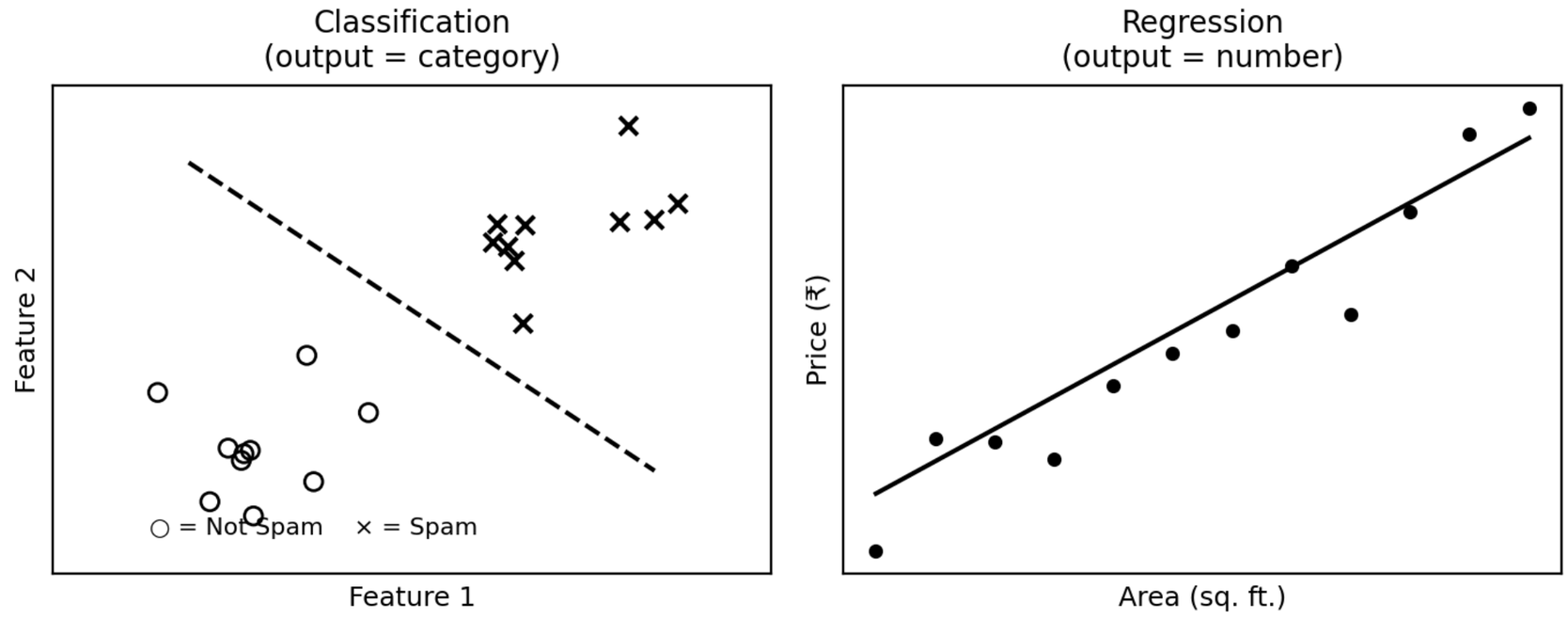



*Fig. 2: Classification (boundary) vs Regression (best-fit line)*

## Question 3 (5 marks)

**Explain the concept of Linear Regression. Write the equation of a simple linear regression model and define the terms (dependent variable, independent variable, intercept, and slope). What is the role of the residual/error term?**

### 1. Concept

Linear Regression is a supervised algorithm that models the relationship between input(s) and a **continuous output** by fitting the **best straight line** through the data.

## 2. Equation (Simple Linear Regression)

$y = \beta_0 + \beta_1 x + \varepsilon$ (also written $y = mx + c$)

## 3. Terms

- **Dependent variable (y):** the output we predict (e.g., marks).
- **Independent variable (x):** the input/predictor (e.g., hours studied).
- **Intercept ($\beta_0$):** value of y when x = 0 (where the line cuts the y-axis).
- **Slope ($\beta_1$):** change in y for a one-unit increase in x.

## 4. Role of Residual / Error Term

Residual **e = actual y – predicted $\hat{y}$**. It captures noise and factors missing from the model, which is why points do not lie exactly on the line. The best line is found by **Least Squares**, which minimizes $\Sigma e^2$.

## 5. Example

Marks = 30 + 10 × Hours. For 5 hours: predicted = 30 + 50 = **80**. If actual marks are 84, residual = 84 – 80 = **+4**.

## 6. Using Residuals

Small, random residuals → good fit. Residuals showing a pattern → the straight-line model is not suitable.

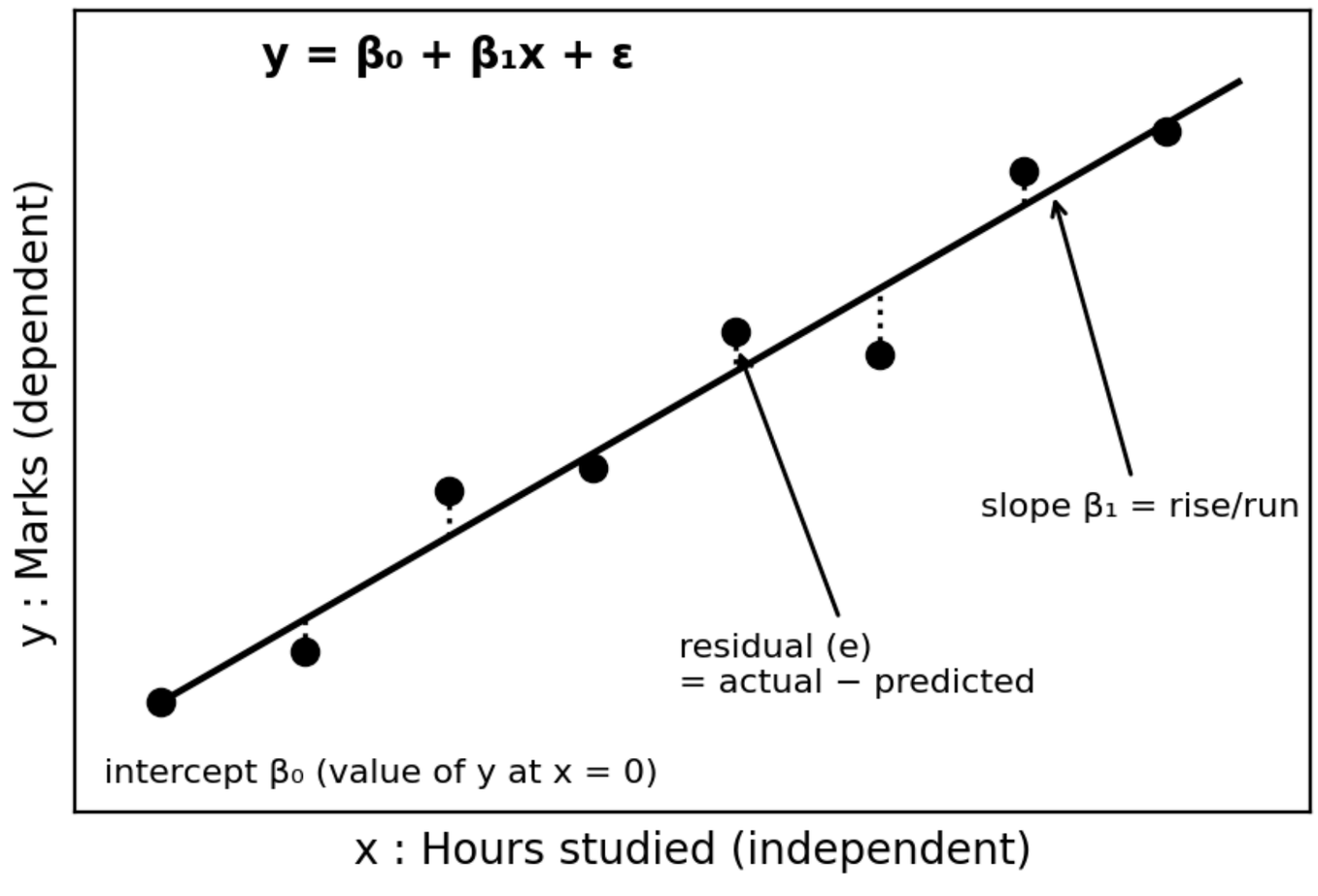



*Fig. 3: Regression line, intercept, slope and residuals (dotted)*

# Question 4 (5 marks)

**How does Logistic Regression differ from Linear Regression? Write the logistic (sigmoid) function and explain why it is suitable for binary classification. What is the decision boundary in logistic regression?**

## 1. Difference from Linear Regression

| Basis | Linear Regression | Logistic Regression |
|---|---|---|
| **Output** | Any continuous value | Probability between 0 and 1 |
| **Problem type** | Regression | Classification |
| **Curve** | Straight line | S-shaped (sigmoid) curve |

| Loss function | Mean Squared Error | Log-loss (cross-entropy) |
|---|---|---|
| Example | Predict house price | Predict pass / fail |

## 2. Sigmoid Function

$\sigma(z) = 1 / (1 + e^{-z})$, where $z = \beta_0 + \beta_1 x_1 + \ldots$ The output always lies between 0 and 1.

## 3. Why Suitable for Binary Classification

It squashes any real number into 0–1, so the output can be read as a **probability**. If $p \geq 0.5 \rightarrow$ Class 1, else Class 0. Large positive z gives p near 1; large negative z gives p near 0.

## 4. Decision Boundary

The boundary is where **p = 0.5**, i.e., **z = 0**, so $\beta_0 + \beta_1 x = 0$. It is a line (or plane) that separates the two classes. The 0.5 threshold can be changed.

## 5. Example

Let $z = -4 + 1 \times$ Hours. Boundary: Hours = 4. For 6 hours: $z = 2$, $p = \sigma(2) = 0.88 \rightarrow$ **Pass**. For 2 hours: $z = -2$, $p = 0.12 \rightarrow$ **Fail**.

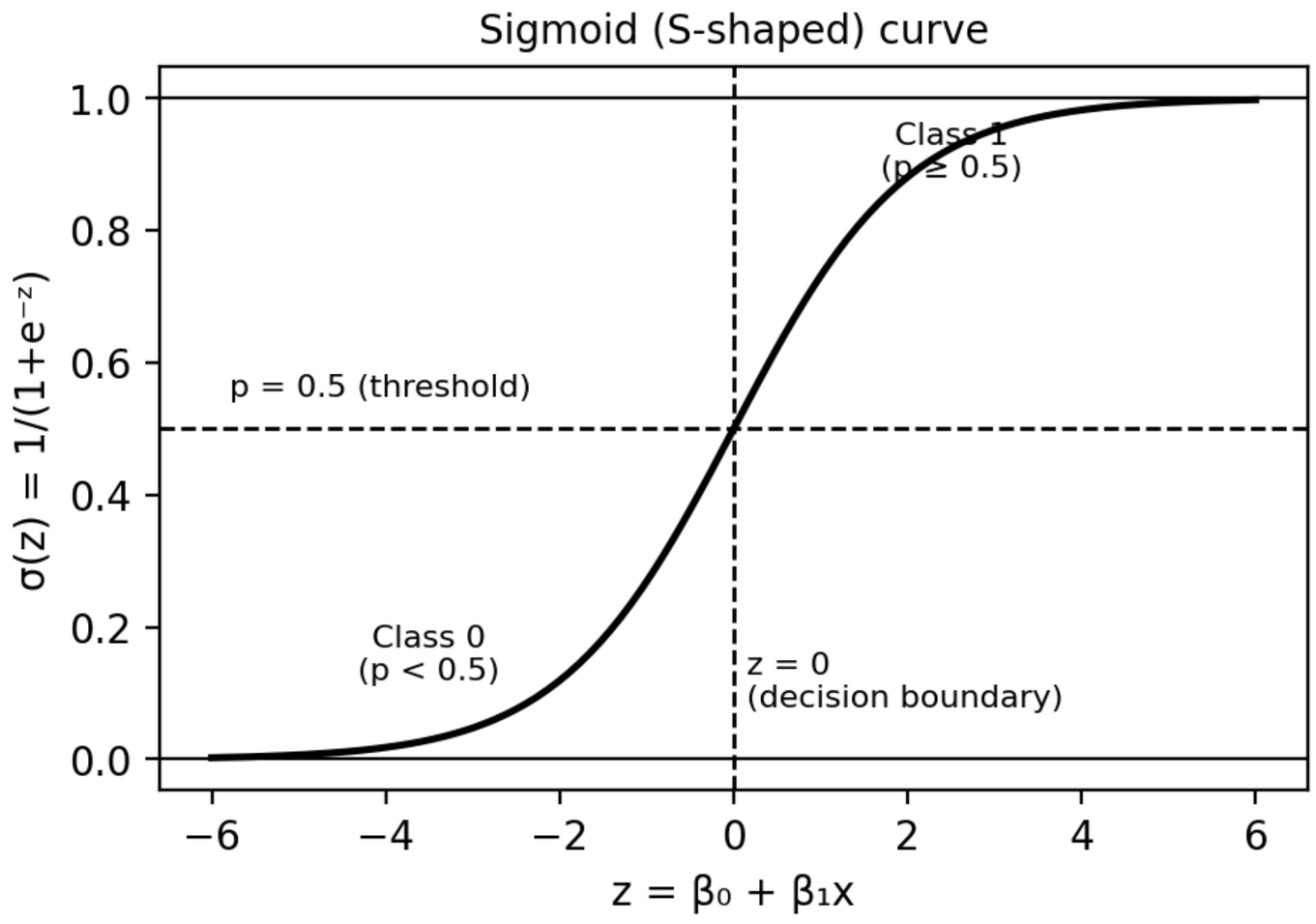



*Fig. 4: Sigmoid curve with threshold p = 0.5*

## Question 5 (5 marks)

**Describe how a Decision Tree works for classification. Explain the concepts of Information Gain and Gini Impurity. Why can decision trees overfit the training data, and how can this problem be reduced?**

## 1. How a Decision Tree Works

It is a flowchart-like model. The **root node** asks a question on a feature, **branches** show the answers, and **leaf nodes** give the class label. Data is split repeatedly on the best feature until nodes are pure. A new sample follows one path from root to leaf.

## 2. Entropy & Information Gain

**Entropy** = $-\Sigma\ p_i\ \log_2\ p_i$ measures disorder (0 = pure, 1 = 50/50 split). **Information Gain = Entropy(parent) − weighted Entropy(children)**. The feature with the **highest gain** is chosen for the split (ID3, C4.5).

### 3. Gini Impurity

**Gini = 1 – Σ $p_i^2$** (0 = pure, 0.5 = worst for two classes). A **lower** Gini is better; it is used in CART. Example: a node with 5 Yes and 5 No → Gini = 1 – (0.25 + 0.25) = **0.5**.

### 4. Why Trees Overfit

A tree can keep growing until every leaf is perfectly pure. It then **memorizes noise and outliers**, giving high training accuracy but poor test accuracy (high variance).

### 5. How to Reduce Overfitting

- **Pre-pruning:** limit max depth, minimum samples per split/leaf.
- **Post-pruning:** grow the full tree, then cut weak branches.
- Use **Random Forest** (many trees) and cross-validation.

### 6. Example – Loan Approval

Income > ₹30,000? Yes → Has existing loan? No → **Approve**; Yes → Reject. Income not above ₹30,000 → **Reject**.

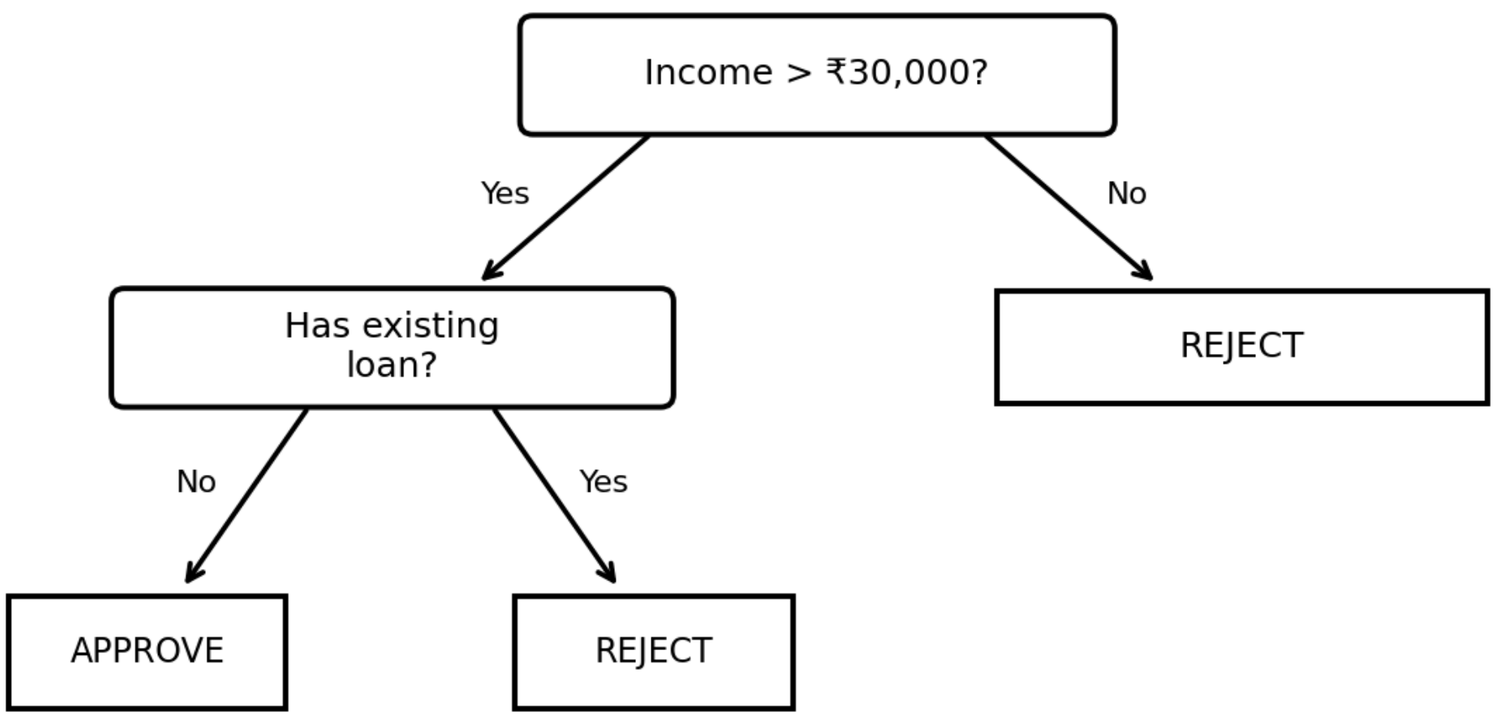



*Fig. 5: Decision tree flowchart (loan approval)*

## Question 6 (5 marks)

**What are Ensemble Methods? Differentiate between Bagging and Boosting with suitable examples of algorithms that use each technique. Why do ensemble methods generally perform better than a single model?**

### 1. Ensemble Methods

Ensemble methods **combine several models** (weak learners) to build one stronger model. Idea: "wisdom of the crowd", as many opinions beat one.

### 2. Bagging (Bootstrap Aggregating)

Models are trained **in parallel** on random samples drawn with replacement. Results are combined by **majority vote** (classification) or **average** (regression). It mainly **reduces variance**. Example: **Random Forest**.

### 3. Boosting

Models are trained **one after another**; each new model focuses on the **mistakes** of the previous one. It mainly **reduces bias**. Examples: **AdaBoost, Gradient Boosting, XGBoost**.

## 4. Bagging vs Boosting

| Basis | Bagging | Boosting |
|---|---|---|
| **Training** | Parallel, independent | Sequential, dependent |
| **Data** | Random bootstrap samples | Re-weighted data (errors get more weight) |
| **Main aim** | Reduce variance (overfitting) | Reduce bias (underfitting) |
| **Model weight** | Equal vote | Better models get more weight |
| **Algorithms** | Random Forest | AdaBoost, XGBoost |

## 5. Why Ensembles Perform Better

Different models make **different errors**, and these cancel out when combined. This gives lower variance/bias, more stability and less overfitting.

## 6. Example

Three independent models, each 70% accurate. Majority vote is right when at least 2 are right: $0.7^3 + 3(0.7^2)(0.3) = 0.343 + 0.441 = \mathbf{0.784} \approx \mathbf{78\%}$, which is better than 70%.

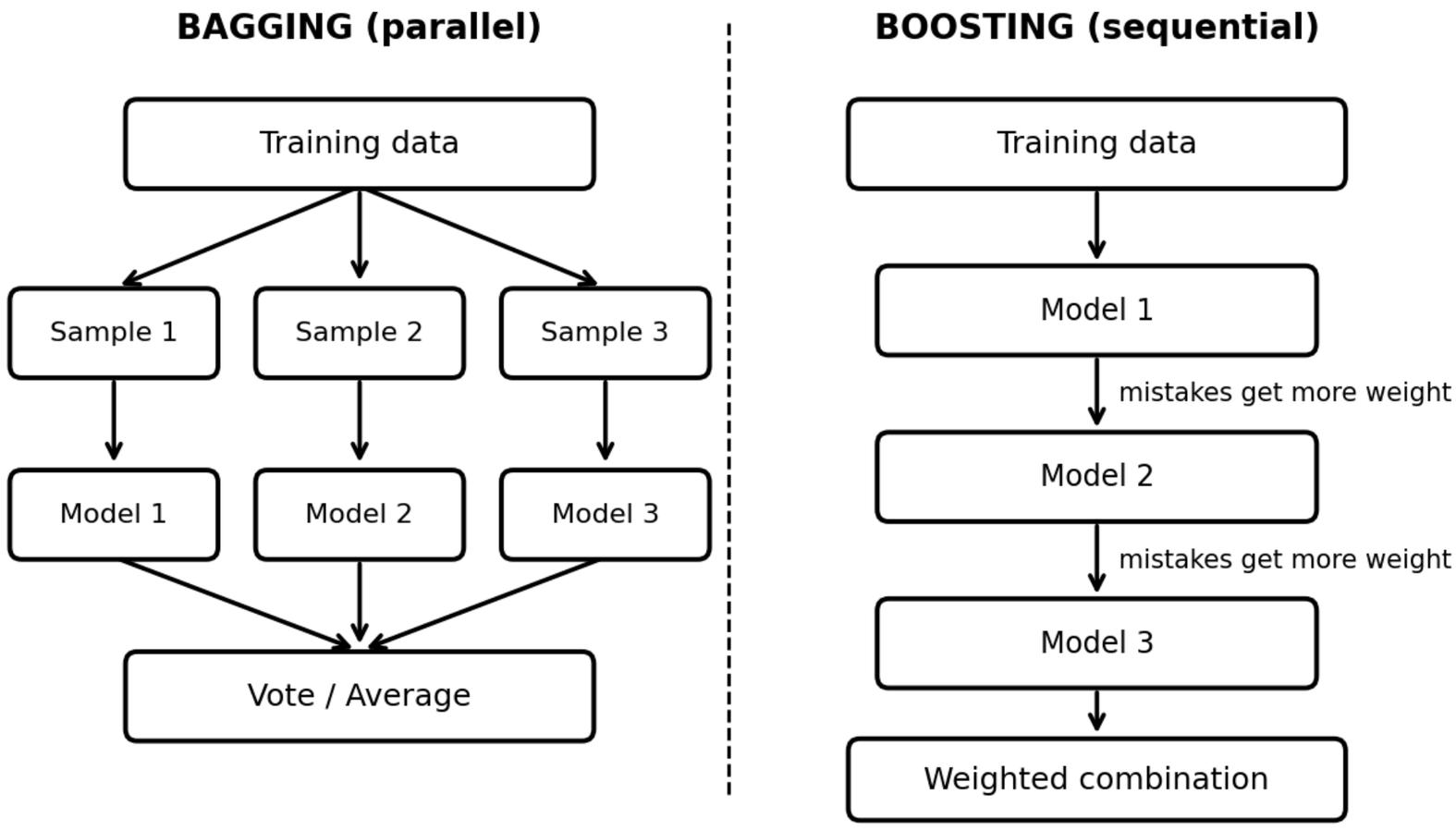



*Fig. 6: Bagging (parallel) vs Boosting (sequential) flowchart*

# Question 7 (5 marks)

**Explain the working principle of Support Vector Machines (SVM) for binary classification. What are support vectors and the margin? Briefly describe the role of the kernel trick in SVM.**

## 1. Working Principle

SVM finds a **hyperplane** (a line in 2-D) that separates the two classes: $\mathbf{w \cdot x + b = 0}$. Points on one side are Class +1 and on the other side Class −1.

## 2. Best Hyperplane = Maximum Margin

Many lines can separate the classes. SVM picks the one with the **largest gap** to the nearest points of both classes, which generalizes better to new data.

## 3. Margin

The margin is the distance between the two dashed boundary lines ($w \cdot x + b = \pm 1$). **Margin = 2 / ||w||,**

so maximizing the margin means minimizing ||w||.

### 4. Support Vectors

They are the **data points closest to the hyperplane**, lying on the margin boundaries. Only they define the hyperplane; removing any other point changes nothing.

### 5. Kernel Trick

When data is **not linearly separable**, a kernel computes similarity as if the data were mapped to a **higher dimension**, without actually doing the mapping. Common kernels: Linear, Polynomial, **RBF (Gaussian)**.

### 6. Example

1-D data: × at both ends and ○ in the middle cannot be split by one cut. Map $x \rightarrow (x, x^2)$ and a **straight line** now separates them.

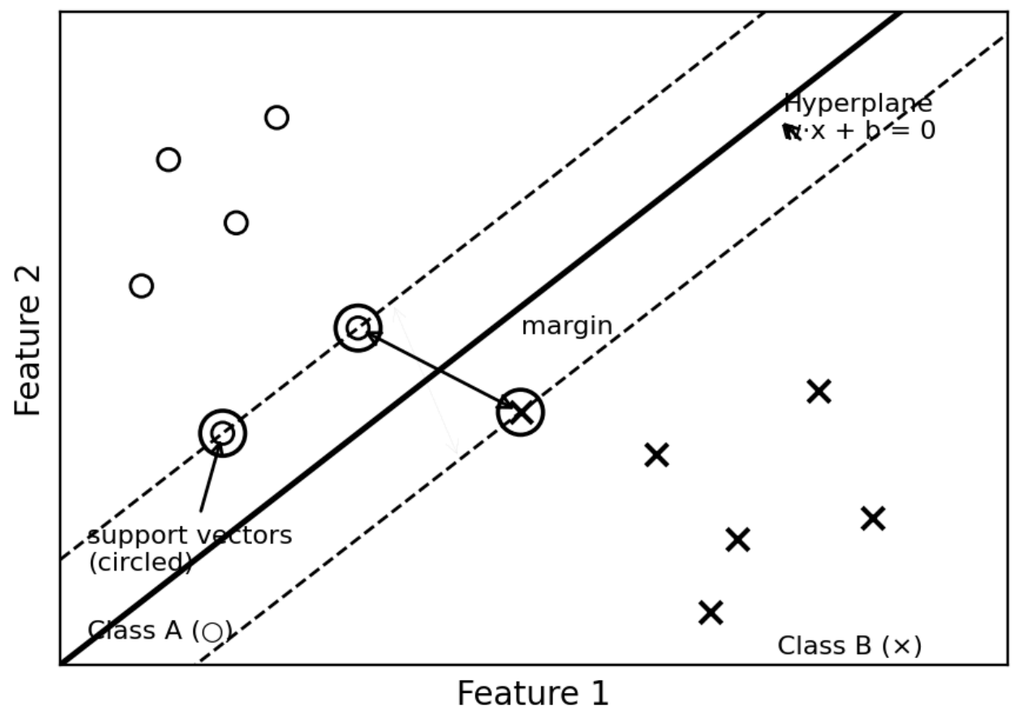



*Fig. 7a: Hyperplane, margin and support vectors*

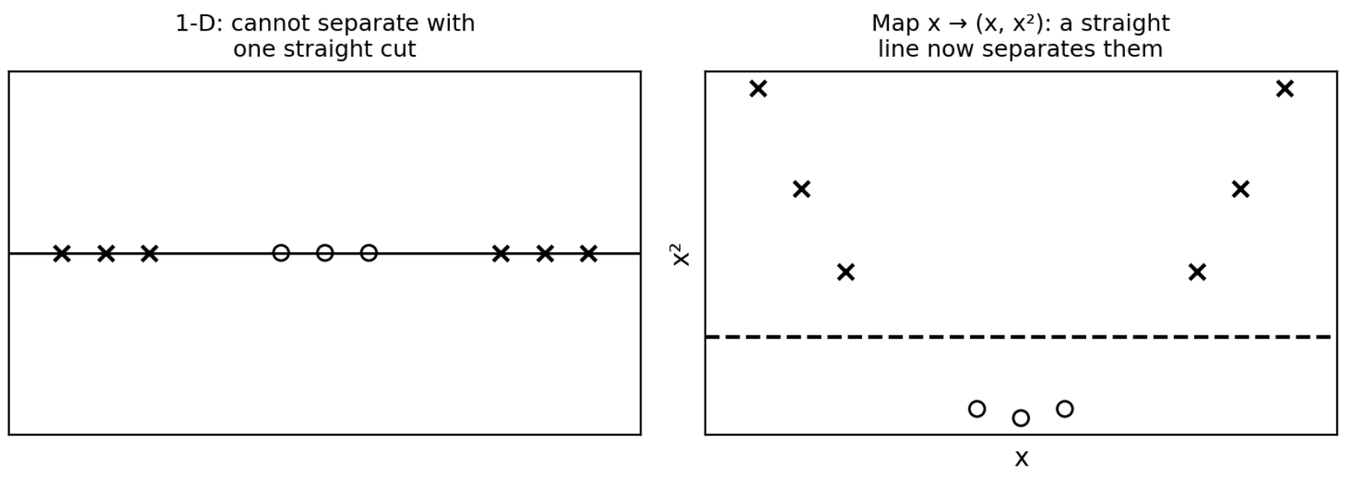



*Fig. 7b: Kernel idea, mapping 1-D data to 2-D*

## Question 8 (5 marks)

**List and briefly explain any four-evaluation metrics used for classification problems (e.g., Accuracy, Precision, Recall, F1-Score, ROC-AUC). When is Accuracy not a good metric, and which metric would you prefer instead?**

### 1. Confusion Matrix (Base of All Metrics)

| | Predicted Positive | Predicted Negative |
|---|---|---|
| **Actual Positive** | TP (True Positive) | FN (False Negative) |
| **Actual Negative** | FP (False Positive) | TN (True Negative) |

### 2. Four Metrics

| Metric | Formula | Meaning |
|---|---|---|
| Accuracy | (TP + TN) / (TP + TN + FP + FN) | Fraction of all predictions that are correct |
| Precision | TP / (TP + FP) | Of predicted positives, how many are truly positive |
| Recall | TP / (TP + FN) | Of actual positives, how many were found |
| F1-Score | 2 × P × R / (P + R) | Harmonic mean of Precision and Recall (balance) |

### 3. Worked Example

TP = 40, FN = 20, FP = 10, TN = 930 (1000 samples). Accuracy = 970/1000 = **97%**. Precision = 40/50 = **0.80**. Recall = 40/60 = **0.67**. F1 = 2(0.80)(0.67)/(0.80 + 0.67) = **0.73**.

### 4. ROC-AUC (Bonus Metric)

The ROC curve plots **TPR (Recall) vs FPR** at every threshold. **AUC** is the area under it: 1.0 = perfect, 0.5 = random guessing.

### 5. When Accuracy Fails

Accuracy misleads on **imbalanced data**. Example: 1000 patients, only 10 sick. A model that says "healthy" for everyone scores **99% accuracy** but **0% recall**, as it misses every sick patient.

### 6. Preferred Metric

- **Recall** when a missed positive is costly (cancer, fraud).
- **Precision** when a false alarm is costly (spam filter).
- **F1-Score** or **ROC-AUC** for imbalanced data in general.

## Question 9 (5 marks)

**For regression problems, explain the following evaluation metrics: (a) Mean Squared Error (MSE) (b) Root Mean Squared Error (RMSE) (c) Mean Absolute Error (MAE) (d) R-squared (Coefficient of Determination). Which metric is more sensitive to outliers and why?**

### 1. (a) MSE – Mean Squared Error

**MSE** = $(1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$. Average of squared errors. Always ≥ 0; the lower the better. Squaring punishes large errors heavily. Units are squared.

### 2. (b) RMSE – Root Mean Squared Error

**RMSE** = $\sqrt{MSE}$. It brings the error back to the **same unit as y**, so it is easier to interpret (e.g., "off by ₹1.7 lakh on average").

### 3. (c) MAE – Mean Absolute Error

**MAE** = $(1/n)\ \Sigma\ |y_i - \hat{y}_i|$. Average size of the error, with no squaring. Same unit as y and easy to understand.

### 4. (d) $R^2$ – Coefficient of Determination

$R^2 = 1 - SS_{res} / SS_{tot}$, where $SS_{res} = \Sigma(y - \hat{y})^2$ and $SS_{tot} = \Sigma(y - \bar{y})^2$. It shows the **proportion of variance explained** by the model: 1 = perfect, 0 = no better than predicting the mean.

### 5. Worked Example

Actual = 3, 5, 7, 9; Predicted = 2, 5, 8, 12. Errors = 1, 0, −1, −3. **MSE** = (1 + 0 + 1 + 9)/4 = **2.75**. **RMSE** = $\sqrt{2.75}$ = **1.66**. **MAE** = (1 + 0 + 1 + 3)/4 = **1.25**. $R^2$ = 1 − 11/20 = **0.45** ($\bar{y}$ = 6, $SS_{tot}$ = 20).

### 6. Most Sensitive to Outliers

**MSE (and RMSE)**, because errors are **squared**. In the example, the error of 3 is 60% of the total

absolute error (MAE) but about **82%** of the total squared error (9 out of 11).

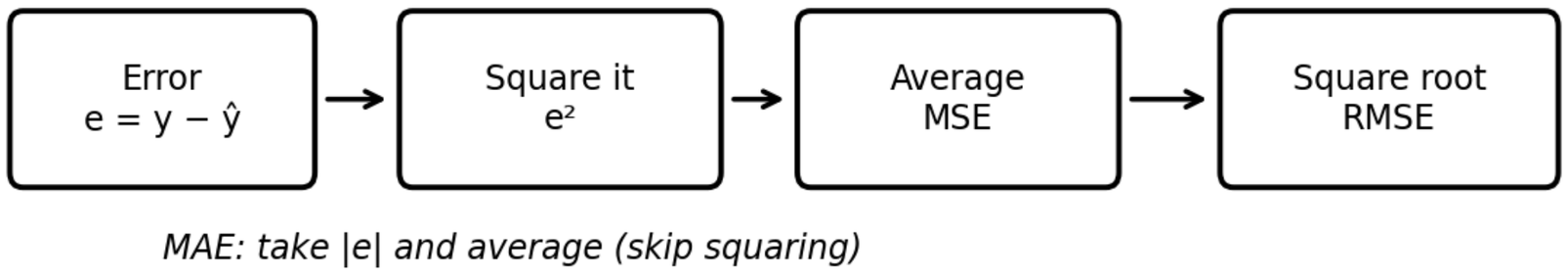



*Fig. 9: How MSE, RMSE and MAE are calculated (flowchart)*

## Question 10 (5 marks)

**Discuss any four real-world applications of Supervised Learning. For each application, mention the type of supervised learning problem (classification or regression) and a suitable algorithm that can be used.**

### 1. Common Idea

All applications follow the same pipeline: learn from **past labeled data**, test the model, then predict for new data.

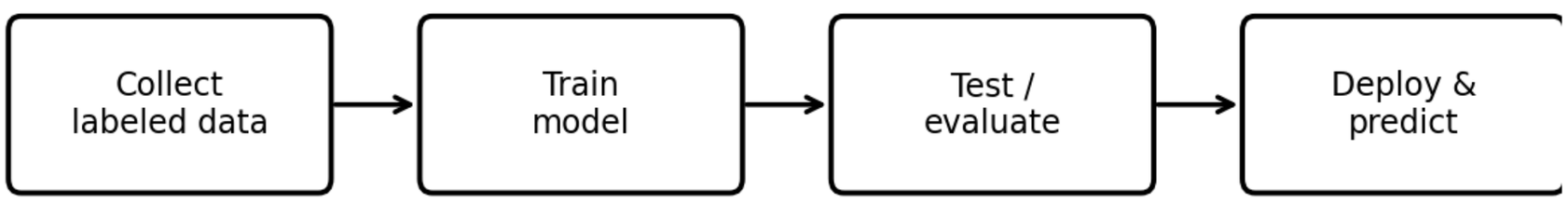



*Fig. 10: General supervised learning pipeline*

### 2. Email Spam Detection

Input: email text/words. Output: Spam or Not Spam. **Type: Classification. Algorithm:** Naive Bayes / Logistic Regression.

### 3. House Price Prediction

Input: area, bedrooms, location. Output: price in ₹. **Type: Regression. Algorithm:** Linear Regression / Random Forest Regressor.

### 4. Medical Disease Diagnosis

Input: patient reports and scan features. Output: Diseased or Healthy. **Type: Classification. Algorithm:** SVM / Decision Tree / Random Forest.

### 5. Retail Sales Forecasting

Input: past sales, season, festivals, price. Output: next month's sales amount. **Type: Regression. Algorithm:** Gradient Boosting / Linear Regression.

### 6. Summary

| Application | Output | Problem Type | Algorithm |
|---|---|---|---|
| **Spam detection** | Spam / Not Spam | Classification | Naive Bayes, Logistic Regression |
| **House price** | Price (number) | Regression | Linear Regression, Random Forest |

| | | | |
|---|---|---|---|
| **Disease diagnosis** | Diseased / Healthy | Classification | SVM, Decision Tree |
| **Sales forecasting** | Sales (number) | Regression | Gradient Boosting |